# LEVEL 2 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

MARK SCHEME

Sample Assessment

**Paper: RFSML2SAM01**

**Functional Skills in Mathematics Level 2 – Mark scheme**

**Paper: RFSML2SAM01**

| Task 1 NC | Process (Task description) | Total mark | Mark allocation | Comments | PS or US | Subject content |
|---|---|---|---|---|---|---|
| **Question 1** | Correct addition of fractions | 2 | **1 mark:** Correct addition of two or more fractions or mixed numbers, eg<br>1½ + ¾ = 2¼ | | US | 7b |
| | Correct mileage | | **1 mark:** Calculate total mileage ie $4\frac{1}{6}$ miles | Accept 4.16, 4.17 | US | 7b |
| **Question 2** | Correct order | 1 | **1 mark:** 3/8, 5/8, ¾, 7/6, 4/3 | Do not accept largest to smallest. Accept 1 1/6 and 1 1/3. | US | 7a |
| **Question 3** | Correct division | 1 | **1 mark:**<br>273696 ÷ 24 = 11404 | | US | 2 |
| **Question 4** | Use formula to calculate surface area | 2 | **1 mark:** 15 x 15 = (225)<br>225 x 6 = (1350) | | US | 17b |
| | Correct answer with units | | **1 mark:** 1350cm$^2$ | Must show units | US | 17b |
| **Question 5a** | Use scale accurately | 2 | **1 mark:** Valid method to calculate length, eg<br>7.5 x 1500 = (11250) OR<br>1.5 x 7.5 = (11.25) OR<br>Other valid method | May be implied if 11.25 seen | PS | 18a |
| | Correct length in metres | | **1 mark:** correct length shown ie 11.25 (m) | Units not required | PS | 18a |
| **Question 5b** | Method to find area of patio | 3 | **2 marks:** Valid method to find the area of the trapezium eg<br>½ (8.4 + 6.6) × 4 = (3 0) OR<br>(8.4 × 4) – (1/2 x 1.8 × 4) OR<br>(6.6 × 4) + (1/2 x 1.8 × 4) OR<br>Other valid method | Award 1 mark for correct area of triangle, 3.6m$^2$ | PS | 16b |
| | Correct area of patio | | **1 mark:** Overall area of patio, ie<br>30m$^2$ | Units required | PS | 16b |
| **Question 5c** | Calculate amount of dry mixture | 4 | **1 mark:** Calculate total amount of dry mixture required, eg<br>30 × 20kg = 600kg. | Allow FT for their area. | PS | 11a |
| | Understanding of ratio shown | | **1 mark:** Evidence of understanding of correct use of ratio, eg | Award if 3.33 seen<br>Award if 100 seen | PS | 11a |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | 1 in 6 OR 1/6$^{th}$ OR 6 parts seen OR 20/6 OR other valid calculations of ratio. | | | |
| | Method to calculate number of bags of cement | | **1 mark:** Method to calculate no of bags of cement, eg<br>(600 × 1/6) ÷ 25 OR<br>600 ÷ 6 ÷ 25 OR<br>3.33 × 30 AND 99.99 ÷ 125<br>OR<br>equivalent valid calculation. | Allow FT for their amount of dry mix. | PS | 11a |
| | Correct number of bags of cement | | **1 mark:** Correct answer, ie<br>4 bags. | Allow FT for their amount of dry mix | PS | 11a |

| Task 2 | Process (Task description) | Total mark | Mark allocation | Comments | PS or US | Subject content |
|---|---|---|---|---|---|---|
| **Question 6** | Calculate total budget for house | 5 | **1 mark:** ((28 145 x 3.5) + 4 875) = (£)103 382.50 | Accept 103 382.5 | PS | 2 |
| | Method to convert distance | | **1 mark:** 0.6 (m) x 1.6 = (0.96 km) | Accept any valid method to convert distance from miles to km<br><br>Implied if 0.96 seen | PS | 14a |
| | Interpret scatter graph | | **1 mark:** Identify cost of available house at required distance from station = | Allow between 105 000 and 107 000<br><br>Award mark if implied by explanation. | PS | 28b |
| | Correct final answer and reason | | **1 mark:** No (with valid calculations)<br><br>**1 mark:** for valid reason, eg because he needs £105 000 but he can only afford £103 382.50 | Accept second mark for reason on FT if a correct reason is given based on their calculations. | PS | 28b |
| **Question 7** | Find the mode | 1 | **1 mark**: Correct mode, ie 11 | | US | 23b |
| **Question 8** | List in order of size | 2 | **1 mark**: Correct order ie:<br>9 9.5 10 10.5 11 12 15 23 | | US | 23a |
| | Correct median | | **1 mark**: Correct median, ie 10.75. | | US | 23a |
| **Question 9** | Calculate time taken to walk | 2 | **1 mark:** Correct calculation of the time to walk to the station, eg<br>2 miles at 3mph = 2 ÷ 3 × 60 = 40 mins | Accept 0.66 hours. | PS | 15a |
| | Correct time for leaving house | | **1 mark:** Correct time to leave home, ie 9.22(am) | | PS | 15a |
| **Question 10** | Method to calculate compound interest | 5 | **1 mark**: Correct calculation of interest 1.75% of £8500 eg<br>0.175 × 8500 = (£)148.75 for Money Saver | Award if 8648.75 or 8954.10 seen | PS | 13a |
| | Correct interest after 3 years for Money Saver | | **2 marks**: Correct calculation for compound interest used to find Money Saver balance after 3 years, eg<br>Correct amount after 1 year ie 8500 + 148.75 = (£)8648.75 then<br>Correct amount after 2 years ie 8648.75 + 151.35 = (£)8800.10 then<br>Correct amount after 3 years ie 8800.10 + 154.00 = (£)8954.10 | Award 1 mark for correct balance of Money Saver account after 2 years.<br><br>Award 2 marks if 8954.10 seen.<br><br>Award 1 mark for correct method. | PS<br>PS | 13a<br>13a |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | Allow FT for their interest.<br>Units not required. | | |
| | Correct interest for Bonus Saver<br>Difference in total balances | | **1 mark:** Correct answer for Bonus Saver ie (£)8946.25 | Units not required | PS | 13b |
| | | | **1 mark**: £7.85 | | PS | 13a |
| **Task 3** | **Process (Task description)** | **Total mark** | **Mark allocation** | **Comments** | **PS or US** | **Subject content** |
| **Question 11** | Correct substitution | 3 | **1 mark**: Correct substitution into formula. | | US | 3 |
| | Correct answer to part in brackets | | **1 mark**:<br>0.2 OR<br>1/5 OR<br>1/25 seen. | May be implied if 4 seen | US | 12 |
| | Correct % given | | **1 mark**: 4 | % sign not required | US | 12 |
| **Question 12a** | Method to calculate sun hours in 2017 | 3 | **1 mark:** Valid method to calculate 2017 sun hours from the given mean, eg 94.5 × 12 months = 1134 | May be implied if 31 seen. | PS | 25 |
| | Find total sun hours except Dec 2017 | | **1 mark:** Add 47 + 61 + 119 + 128 + 214 + 108 + 144 + 126 + 94 + 56 + 6 (= 1103) | May be implied if 31 seen. | PS | 25 |
| | Subtraction | | **1 mark:** 1134 - 1103 =31 OR<br>Other valid calculation method AND<br>'Yes, Raheema is correct' | Do **not** award if 31 not seen. | PS | 25 |
| **Question 12b** | Correct year identified by comparing ranges | 1 | **1 mark:** 2017<br>Eg<br>206 - 21 = 185 AND<br>214 - 6 = 208 | Do **not** award if no supporting calculations of range. | PS | 25 |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Question 12c** | Correct kWh calculated<br><br>Correct kWh per month<br><br>Correct cost of electricity | 3 | **1 mark**: Correct number of kWh ie.<br>1.225 ÷ 1.09 = 1.123853211009174 | Award for rounding to 2 or 3 dp, ie 1.12 OR 1.124 | PS | 10d |
| | | | **1 mark**: Correct number of kWh in June, ie<br>1.123853211009174x1.08 = 121.376146789 | Allow FT from their number of kWh<br>Allow FT for rounded figures, eg<br>1.124 x 108 = 121.392<br>1.12 x 108 = 120.96 | PS | 10c |
| | | | **1 mark:** Correct cost of electricity, ie<br>121.376146789x0.143 = (£) 17.35 OR £17.36 | Allow FT For rounded figures to 2 or 3 dp, eg<br>120.96 x 0.143 = (£)17.29 OR 17.30<br>121.392 x 0.143 = (£)17.36<br>121.4 x 0.143 = 17.36<br><br>Allow for rounding.<br><br>Do not award for more or less than 2 dp. | PS | 10c |
| **Question 12d** | Method to calculate volume<br><br>Correct volume<br><br><br><br>Method to convert volume to gallons<br><br>Correct number of gallons<br><br>Valid explanation given | 5 | **1 mark**: Valid method 3.14 × 0.4 × 0.4 × 1 = (0.5024) | Must be consistent units.<br>Do **not** award for use of diameter. | PS | 17a |
| | | | **1 mark**: Correct answer = 0.5024<br><br>Accept 0.502 – 0.503<br><br>*Can use range of 3.14 to 3.142 for pi.* | May be implied if 0.5024 seen. | PS | 17a |
| | | | **1 mark:** Method to convert volume to gallons, eg = 0.5024 × 219.97 | Allow FT for their volume.<br>May be implied if 110.51 gallons seen. | PS | 14c |
| | | | **1 mark:** Correct number of gallons<br>= 110.51 (gallons) | | PS | 14c |
| | | | **1 mark:** Valid explanation, eg "'Yes, she is correct, the container will hold more than 100 gallons". | Accept other valid answers.<br>Do not accept 'yes' without supporting calculations.<br><br>Allow FT for incorrect volume or number of gallons. | PS | 17a |
| **Task 4** | **Process** | **Total** | **Mark allocation** | **Comments** | **PS or US** | **Subjec** |

| | (Task description) | mark | | | | t content |
|---|---|---|---|---|---|---|
| **Question 13** | Plot coordinate on grid | 1 | **1 mark**: Point plotted correctly on graph | | US | 19 |
| **Question 14** | Calculate the decimal<br><br>Convert to fraction in simplest form | 2 | **1 mark:** correct calculation of decimal, ie (144 ÷ 240 = 0.6) converted to 6/10 | | US | 8 |
| | | | **1 mark**: 3/5 | | US | 8 |
| **Question 15a** | Correct entry fees and percentage<br><br>Calculate the ratio<br><br>Calculate total income<br><br>Calculate total profit | 4 | **1 mark:** Complete entry fees in table, ie £300 and 25% | May be implied if 1200 or 228 or 384 or 108 or 180 seen. | PS | 11b |
| | | | **1 mark:** Find appropriate ratio, ie £:% as 300:25 OR 12:1 or 300 × 4. | May be implied if 1200 or 228 or 384 or 108 or 180 seen. | PS | 11b |
| | | | **1 mark:** Find total income, ie (£) 1200. | Units not required. | PS | 11b |
| | | | **1 mark:** Calculate total profit, ie 1200 – 175 – 85 = (£) 940 | Allow FT using their total income figure.<br>Units not required. | PS | 11b |
| **Question 15b** | Correct circumference<br><br>Correct ribbon length for 15 cakes<br><br>Calculate extra 12.5%<br><br>Rounded length<br><br>Calculate cost | 5 | **1 mark:** Correct circumference of a cake, eg 2 × 80 × 3.14 = 502.4mm, accept 502 - 503mm | May be implied if 502-503 seen. May use metres or cm eg 8cm or 0.08m | PS | 16a |
| | | | **1 mark:** Calculate ribbon length for 15 cakes, ie 502.4 × 15 = 7536mm | Alt method 12.5% first then × 15 | PS | 16a |
| | | | **1 mark:** Calculate 112.5%, eg 7536 × 1.125 OR equivalent = 8478(mm)<br><br>Accept 8475 – 8481(mm). | Award if correct answer seen | PS | 6 |
| | | | **1 mark:** 9(m) required | Units not required.<br><br>Award if correct answer seen | PS | 6 |
| | | | **1 mark:** correct calculation of cost, eg 9(m) × £4.95 = £44.55 | | PS | 6 |
| **Question 15c** | Probability of winning a prize and of spin made by a girl | 3 | **1 mark:** Correct probability of a spin winning a prize given, eg 1/3 OR 4/12<br>**AND** Correct probability of a spin being made by a girl, ie ½ or 0.5 | May be implied if $1/6^{th}$ seen. | PS | 27a |

| | | | | | | |
|---|---|---|---|---|---|---|
| | Method to calculate probability of 2 events<br>Correct probability of 2 events | | **1 mark:** Method to calculate probability of a person being a girl and winning a prize, ie 1/3 × ½ = OR 0.5 × 0.33 = | Allow FT for their two individual probabilities.<br>May be implied if $1/6^{th}$ seen. | PS | 26 |
| | | | **1 mark** - Correct probability of 2 events, ie 1/6 OR 0.166 OR 16.6% | Allow FT for their two individual probabilities. | PS | 26 |

**Annotation notes:**

| Annotation | Meaning |
|---|---|
| US | Underpinning skills |
| PS | Problem solving skills |
| FT | Follow through |
| (…) | Information that is not required for the mark point |